Bike Sharing Demand Forecasting: A Count Regression Analysis using Poisson and Negative Binomial GLM

William Chen

Abstract

This report analyzes daily bike rental counts from the UCI Bike Sharing Dataset, comprising 731 observations collected from the Capital Bikeshare system in Washington D.C. between 2011 and 2012. Since *cnt* is a non-negative integer, Poisson GLM is a more theoretically appropriate choice than OLS. However, the raw Variance/Mean ratio of 833 and a post-fit dispersion estimate of 182.6 reveal severe overdispersion, violating the Poisson assumption of equal mean and variance. Negative Binomial regression (NB2) is adopted to address this, reducing the dispersion estimate to 1.046 and the AIC from 137,807 to 12,193. Zero-Inflated Poisson (ZIP) is found inapplicable as the minimum observed count is 22. The final NB2 model achieves a McFadden R^2 of 0.761 and a rolling-origin CV RMSE of 1,660, and identifies *temp* (IRR = 4.51), *yr1* (IRR = 1.62), and *weathersit3* (IRR = 0.48) as the dominant predictors of daily rental demand.

1 Introduction

Bike-sharing systems have become an increasingly popular form of urban transportation, offering a flexible and environmentally friendly alternative to private vehicles. Accurately forecasting daily rental demand is valuable for operators seeking to optimize fleet distribution, staffing, and infrastructure planning. Demand is inherently influenced by a combination of weather conditions, seasonal patterns, and calendar effects, making it a well-suited problem for regression modeling.

This report analyzes daily bike rental counts using the UCI Bike Sharing Dataset, which records 731 days of rental activity across 2011 and 2012 in Washington D.C. Since the response variable *cnt* is a non-negative integer, count regression is more theoretically appropriate than Ordinary Least Squares. We begin with Poisson GLM as the canonical model for count data, and address the overdispersion present in the data using Negative Binomial regression (NB2). Zero-Inflated Poisson (ZIP) is also considered but found inapplicable given the absence of zero counts in the dataset. Model performance is evaluated using AIC, BIC, dispersion estimate, and McFadden R^2, with out-of-sample predictive performance assessed via rolling-origin cross-validated RMSE and MAE. Coefficients are interpreted via Incidence Rate Ratios (IRR $= e^{\hat{\beta}}$).

2 Data

2.1 Dataset Overview

The dataset contains 731 daily observations collected between January 2011 and December 2012 from the Capital Bikeshare system. It comprises 16 variables, including temporal indicators, weather measurements, and rental usage counts. No missing values were found. The target variable *cnt* represents the total number of bike rentals per day, combining both casual and registered users.

2.2 Feature Description

The dataset contains 16 variables: 2 identifiers (*instant*, *dteday*, excluded from analysis), 1 continuous target (*cnt*), and 13 predictors. Features span temporal (*season*, *yr*, *mnth*, *holiday*, *weekday*, *workingday*), weather (*weathersit*, *temp*, *atemp*, *hum*, *windspeed*), and usage (*casual*, *registered*) dimensions.

Of these 13 predictors, 7 are categorical and 6 are continuous. Two (*casual*, *registered*) are excluded from modeling due to target leakage, leaving 11 predictors for analysis.

2.3 Feature Selection and Preprocessing

Prior to modeling, four variables were removed from the dataset. *instant* and *dteday* are excluded as they serve purely as a record index and a date string respectively, carrying no predictive information. *casual* and *registered* are excluded due to target leakage: they are direct components of *cnt* (i.e. *cnt* = *casual* + *registered*), and including them would trivially inflate model performance without providing genuine predictive value. Further variables (*atemp*, *mnth*, *workingday*) are examined and removed in subsequent sections based on correlation analysis and multicollinearity diagnostics.

3 Exploratory Data Analysis

3.1 Feature Distribution

Figure 1 shows the distribution of daily rental counts. Counts range from 22 to approximately 8,700. The distribution is roughly symmetric but exhibits a slight secondary peak around 7,000–7,500, likely reflecting the higher overall demand in 2012 compared to 2011.

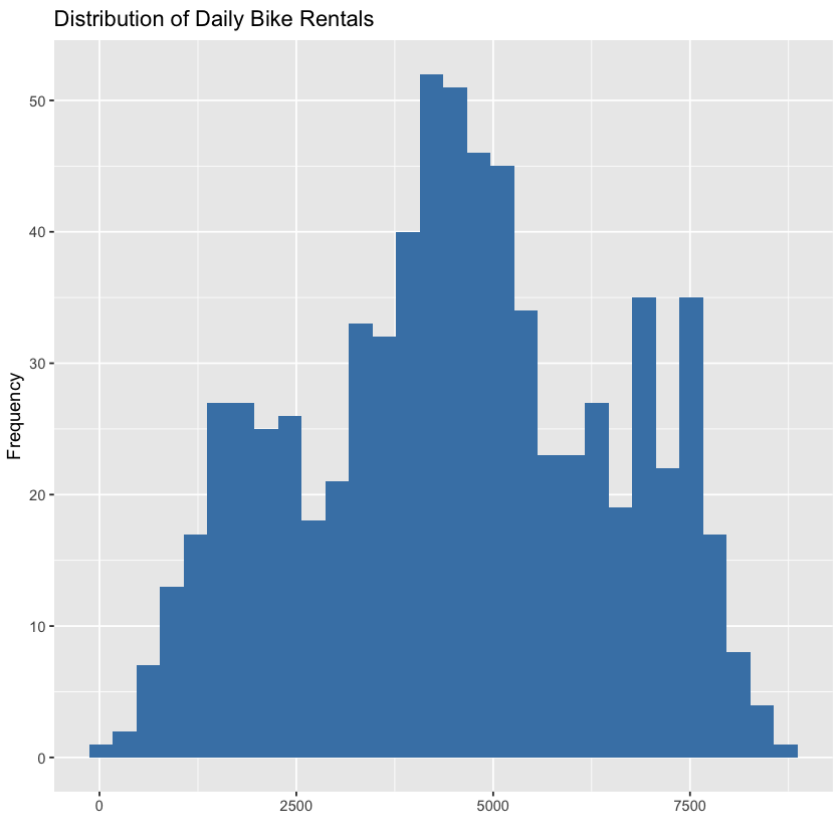

Figure 1: Distribution of Daily Bike Rentals

3.1.1 Categorical Features

Figures 2 and 4 show consistent seasonal trends: counts peak in Summer, followed by Spring and Fall, while Winter records the lowest demand. The two plots convey near-identical distributional patterns, reflecting the strong overlap between the two variables. Figure 3 shows a marked increase in overall demand from 2011 (Year 0) to 2012 (Year 1), with the median rising from approximately 3,500 to 6,200. A small number of outliers are visible across all three plots, representing isolated days with unusually low counts; these are consistent with extreme weather events and are considered a normal occurrence in the data.

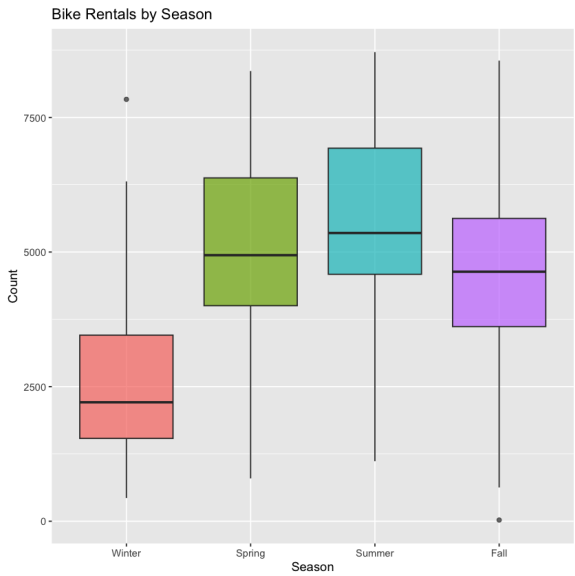

Figure 2: Bike Rentals by Season

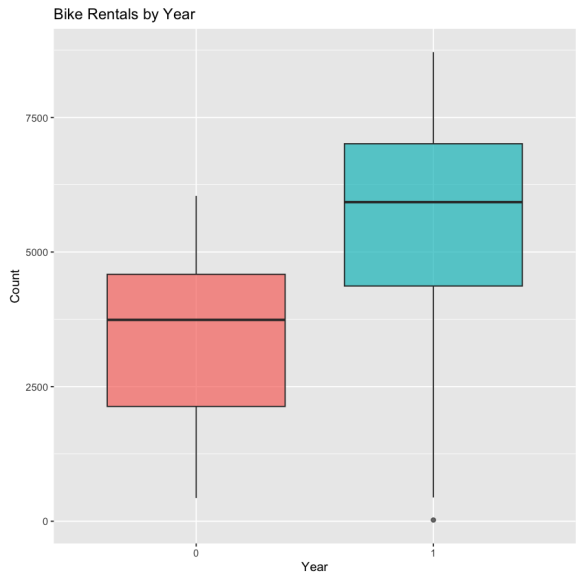

Figure 3: Bike Rentals by Year

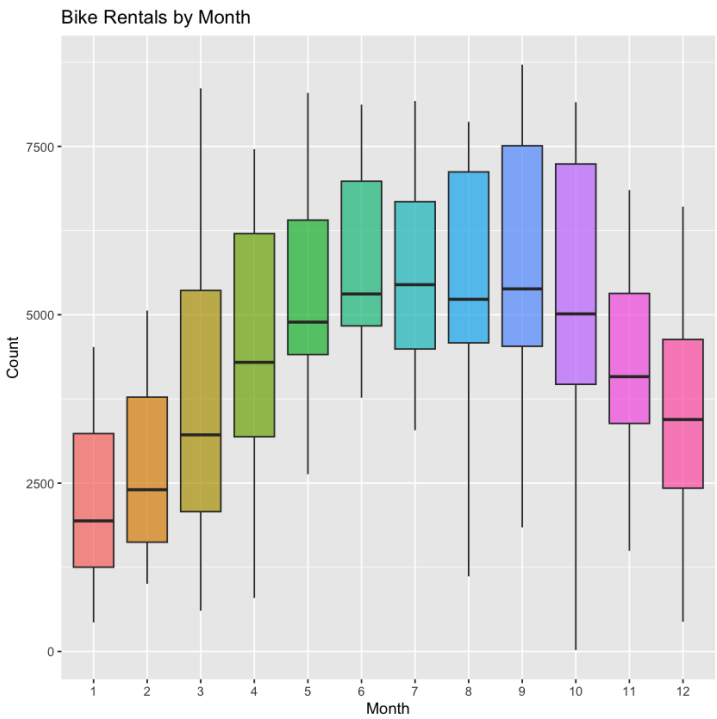

Figure 4: Bike Rentals by Month

Figure 5 shows rental counts across weather conditions and day-type indicators. *weathersit* shows a clear negative trend: higher values indicate more severe weather, and rental counts decline accordingly. Level 4 does not appear in the dataset as no such extreme weather days were recorded. *holiday*, *weekday*, and *workingday* show largely overlapping distributions with no strong separation between categories.

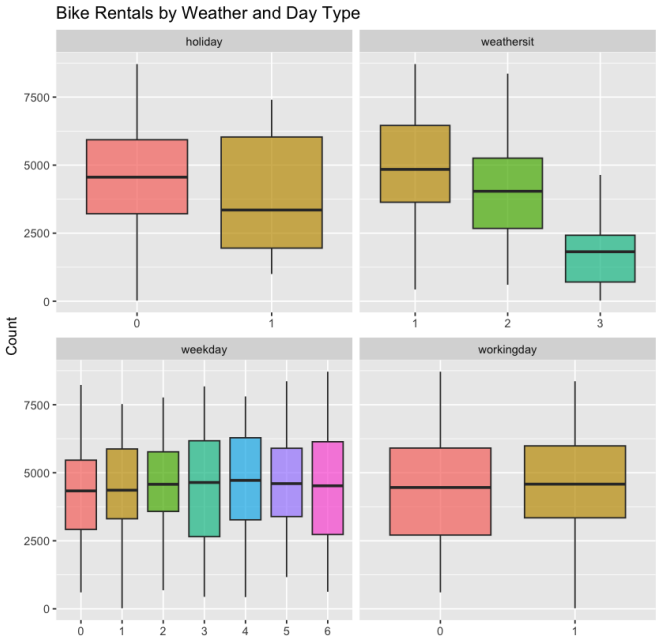

Figure 5: Bike Rentals by Weather and Day Type

3.1.2 Numeric Features

Figure 6 shows the relationship between each numeric feature and daily rental counts. Both *temp* and *atemp* display near-identical positive trends, suggesting that the two variables carry largely redundant information. *hum* and *windspeed* both show negative relationships with rental counts, indicating that higher humidity and stronger winds are associated with fewer rentals.

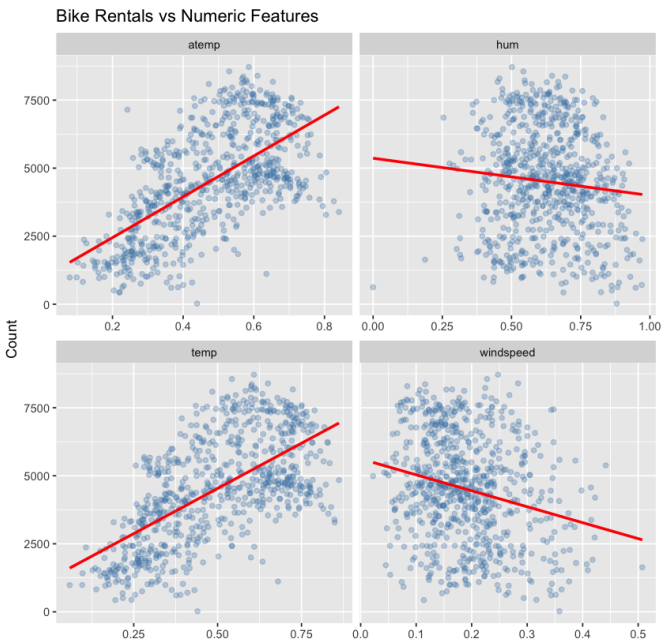

Figure 6: Bike Rentals vs Numeric Features

3.2 Feature Correlation

Figure 7 shows the correlation matrix for numeric features. The most notable finding is the near-perfect correlation between *temp* and *atemp* ($r = 0.99$), confirming the visual observation from the scatter plots and providing a clear justification for removing one of the two variables. Both *hum* ($r = -0.10$) and *windspeed* ($r = -0.23$) show negative correlations with *cnt*, consistent with the trends observed earlier.

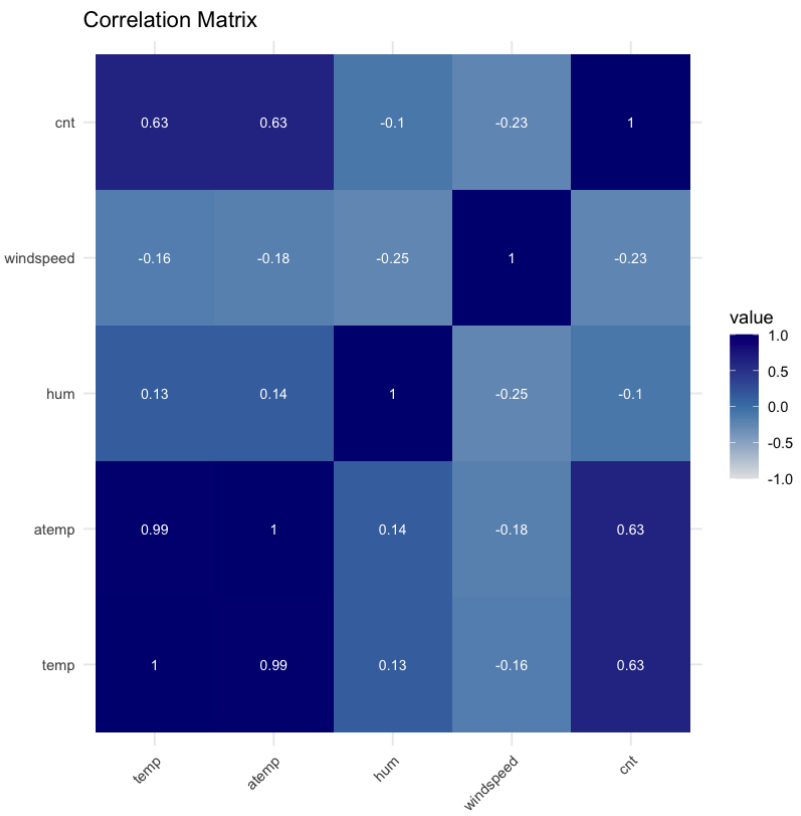

Figure 7: Correlation Matrix of Numeric Features

3.3 Multicollinearity Check (VIF)

Variance Inflation Factor (VIF) was computed to assess multicollinearity among the remaining predictors. Before running VIF, three variables were excluded from the model: *atemp* (near-perfect correlation with *temp*, $r = 0.99$), *mnth* (aliased with *season*), and *workingday* (linearly derived from *weekday* and *holiday*). Including aliased variables causes perfect multicollinearity, making VIF computation impossible due to a singular design matrix.

All remaining features returned VIF values below 3.6, indicating no problematic multicollinearity. The two highest values belong to *season* (3.54) and *temp* (3.41), which reflects a natural association between temperature and season and does not warrant further action.

4 Methodology

4.1 Poisson GLM

Poisson GLM is a generalized linear model suited for count data. It uses a log link function to model the expected count, ensuring all predicted values remain non-negative. The key advantage over OLS is that it respects the discrete, non-negative nature of the response variable. Coefficients are interpreted through the Incidence Rate Ratio (IRR = e^{β}), which represents the multiplicative change in expected count for a one-unit increase in a predictor. A core assumption of Poisson is equidispersion, meaning the variance equals the mean. Violation of this assumption leads to underestimated standard errors and unreliable inference.

4.2 Negative Binomial (NB2)

Negative Binomial regression is an extension of Poisson that introduces an additional dispersion parameter, allowing the variance to exceed the mean. It is the standard approach when overdispersion is detected in count data. Coefficients are interpreted in the same way as Poisson via IRR. When the dispersion parameter α approaches zero, NB2 converges to Poisson, making Poisson a special case of NB2.

4.3 Zero-Inflated Poisson (ZIP)

ZIP is designed for count data with an excess of zero observations. It models zeros as arising from two sources: a structural zero component and a count component. ZIP is considered when the number of zeros in the data exceeds what Poisson or NB2 would predict. In this dataset, the minimum observed count is 22 with no zero counts present, so ZIP is not applicable and is excluded from further analysis.

4.4 Evaluation Metrics

Models are compared using both in-sample and out-of-sample metrics. AIC and BIC balance goodness of fit against model complexity, with lower values indicating a better model. The dispersion estimate (Residual Deviance / degrees of freedom) verifies whether the equidispersion assumption holds, with values close to 1 indicating a well-fitted model. McFadden R^2 serves as a supplementary goodness-of-fit measure; however, it should be interpreted with caution under Poisson when overdispersion is present, as the inflated deviance can produce artificially high values. Out-of-sample performance is assessed using CV RMSE and CV MAE, computed via a rolling-origin strategy to respect the temporal structure of the data. Binned residual plots provide a visual check of model fit by examining whether average residuals fall within the 95% confidence bands.

Quasi-Poisson. An alternative approach for handling overdispersion is Quasi-Poisson, which corrects standard errors by scaling them with the dispersion estimate without changing the coefficients. However, since Quasi-Poisson does not produce a full likelihood, AIC and BIC cannot be computed, making formal model comparison infeasible. Negative Binomial regression is therefore preferred in this analysis.

5 Results

5.1 Poisson GLM

A preliminary dispersion check on the raw data reveals a Variance/Mean ratio of 833.15, indicating severe overdispersion before any model is fitted. After fitting the Poisson GLM, the post-fit dispersion estimate (Residual Deviance / df) drops to 182.6, which still far exceeds 1, confirming that the Poisson assumption of equidispersion is violated even after controlling for all predictors.

Figure 8 shows the binned residual plot for the Poisson model. The majority of points fall outside the 95% confidence bands (grey lines), with residuals ranging from approximately -18 to 9. This visually confirms the poor fit of the Poisson model and motivates the use of Negative Binomial regression.

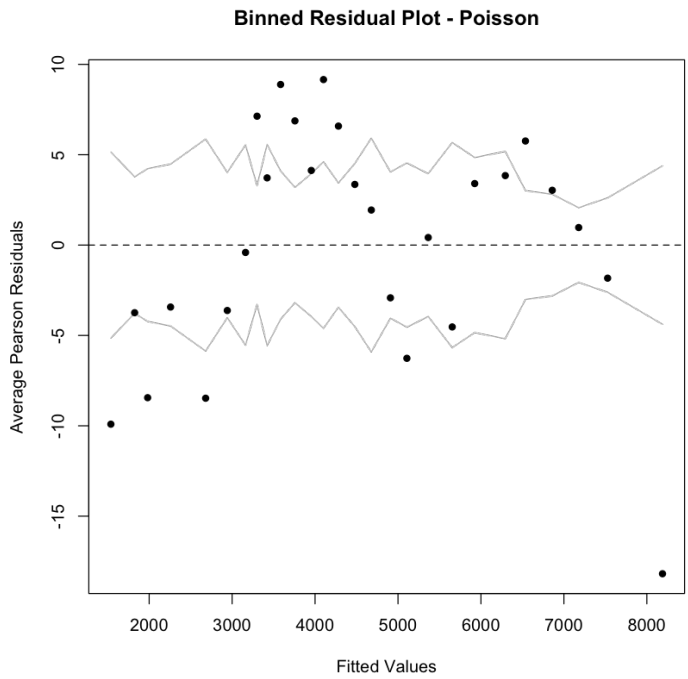

Figure 8: Binned Residual Plot — Poisson GLM

5.2 Negative Binomial (NB2)

The NB2 model yields a dispersion estimate of 1.046 (Residual Deviance 747.07 / df 714), a substantial improvement from the Poisson model's 182.6, indicating that the overdispersion is successfully resolved.

Figure 9 shows the binned residual plot for NB2. Compared to Poisson, the y-axis range narrows dramatically from approximately $[-18, 9]$ to $[-1.0, 0.75]$, and the majority of points fall within the 95% confidence bands. A small number of points near the lower and upper ends of the fitted value range fall slightly outside, suggesting minor difficulties in predicting very low or very high demand days.

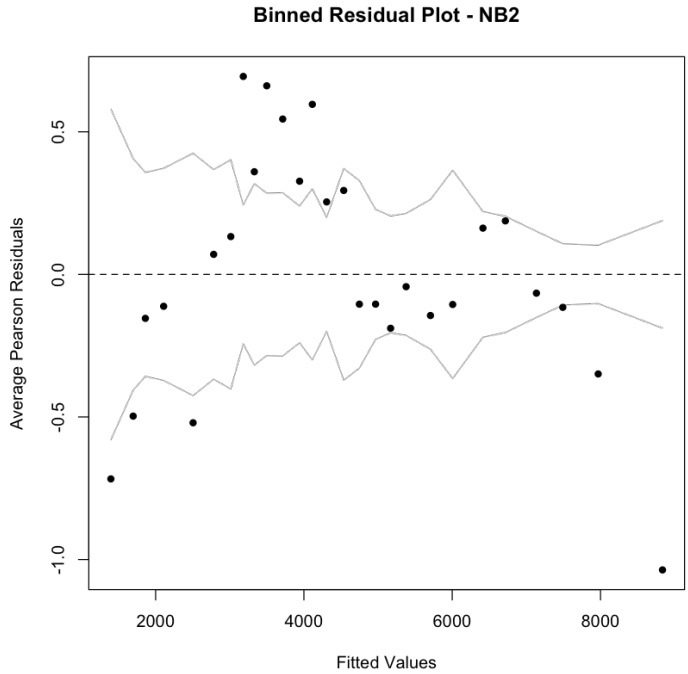

Figure 9: Binned Residual Plot — NB2

Figure 10 shows the actual versus predicted counts. Points align closely along the 45-degree reference line across the full range of demand, confirming good overall predictive performance. Slight deviations are observed at very low and very high counts, reflecting the inherent difficulty of modeling extreme demand days.

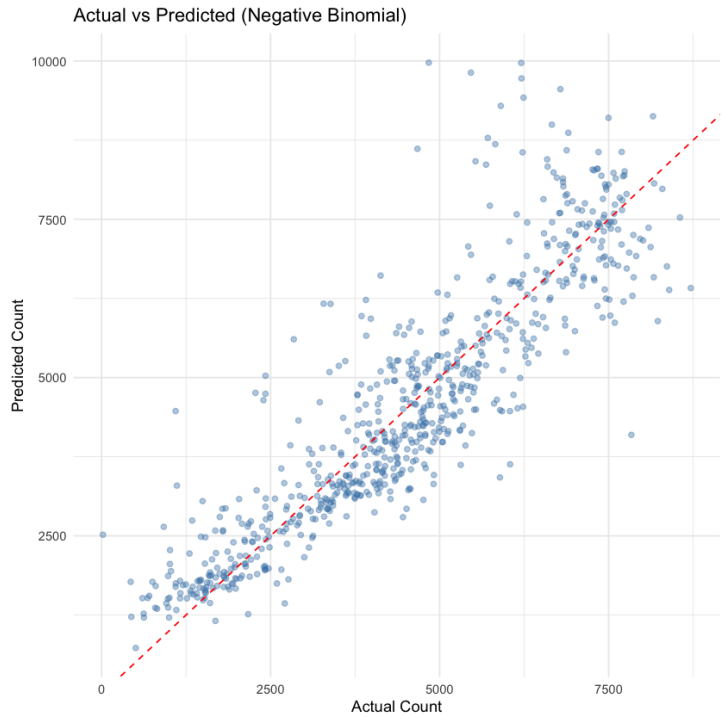

Figure 10: Actual vs Predicted — NB2

Table 1 presents the full set of estimated coefficients, expressed as Incidence Rate Ratios (IRR) with 95% confidence intervals.

Table 1: NB2 Model Coefficients — Incidence Rate Ratios (IRR)

Variable	**IRR**	**95% CI Lower**	**95% CI Upper**
(Intercept)	1593.166	1382.437	1837.075
seasonSpring	1.389	1.301	1.483
seasonSummer	1.249	1.147	1.360
seasonFall	1.616	1.528	1.710
yr1	1.619	1.561	1.678
holiday1	0.809	0.724	0.907
weekday1	1.067	0.996	1.143
weekday2	1.086	1.016	1.161
weekday3	1.075	1.006	1.150
weekday4	1.099	1.028	1.175
weekday5	1.122	1.049	1.200
weekday6	1.094	1.023	1.169
weathersit2	0.902	0.861	0.946
weathersit3	0.483	0.427	0.547
temp	4.508	3.755	5.412
hum	0.744	0.622	0.891
windspeed	0.506	0.393	0.652

Reference categories: Winter (season), Year 0 (yr), Sunday (weekday), Clear weather (weathersit), Non-holiday (holiday). All continuous predictors are normalized to [0, 1].

The results highlight several key patterns. Temperature is the dominant predictor (IRR = 4.508), reflecting strong sensitivity of cycling demand to thermal comfort. Year-over-year growth (IRR = 1.619) and seasonal effects are also substantial, with autumn demand approximately 62% higher than the Winter baseline. Adverse weather (IRR = 0.483) and high wind speed (IRR = 0.506) each suppress demand by roughly half. Among weekday indicators, Monday (*weekday1*) is the only non-significant predictor, suggesting no meaningful difference from the Sunday baseline.

5.3 Model Comparison

Table 2 compares the two models across in-sample and out-of-sample metrics.

Table 2: Model Comparison. AIC, BIC, Dispersion, and McFadden R^2 are in-sample; CV RMSE and CV MAE are out-of-sample, evaluated using a rolling-origin strategy with 5 expanding windows.

Model	**AIC**	**BIC**	**Dispersion**	**McFadden R^2**	**CV RMSE**	**CV MAE**
Poisson	137,806.50	137,884.61	182.596	0.805	1489.626	1143.205
NB2	12,193.16	12,275.86	1.046	0.761	1660.491	1287.878

NB2 outperforms Poisson on AIC, BIC, and dispersion, confirming that overdispersion is resolved. The higher Poisson McFadden R^2 (0.805 vs. 0.761) is an artifact of overdispersion inflating the deviance-based statistic.

6 Discussion

6.1 Key Findings

The NB2 model outperforms the Poisson GLM, as evidenced by lower AIC and BIC and a dispersion estimate close to 1, indicating that the equidispersion assumption is well-satisfied after accounting for the overdispersion present in the raw data. Temperature (IRR = 4.51) is the dominant predictor, with year-over-year growth (IRR = 1.62) and seasonal effects also contributing substantially. Adverse weather and high wind speed suppress demand by roughly half, while most weekday effects are significant except Monday, which does not differ detectably from the Sunday baseline. Although Poisson achieves lower CV RMSE (1489.6 vs. 1660.5), this reflects similar point estimates under the same link function rather than superior model specification; NB2 is preferred for its valid inference under overdispersion.

6.2 Limitations

Data limitations. Two limitations stem from the dataset itself. First, the model covers only two years of data, and the large `yr` coefficient may reflect a growth phase specific to that period rather than a persistent trend, limiting predictive validity for future years. Second, prediction errors are likely larger on extreme weather days, as such events are rare and poorly represented during training.

Methodological limitations. Two limitations concern model specification. First, the model excludes interaction terms such as `temp:season`, which may capture more nuanced demand dynamics and improve performance. Second, the model assumes independence across observations; however, daily rental counts exhibit inherent temporal structure. Residual autocorrelation was not formally assessed in this analysis. Future work could examine ACF plots or apply a Durbin-Watson test on model residuals, and consider time-series extensions such as ARIMA-GLM or generalized additive models with temporal smoothing terms.

References

[1] Fanaee-T, H. (2013). *Bike Sharing* [Dataset]. UCI Machine Learning Repository. `https://doi.org/10.24432/C5W894`